

Abstract Algebra 1 - Homework 2- M. Vicaría

October 15, 2025

Dear student please submit the complete solution to each of the following problems in Gradescope in a PDF file. Please write down your name in each page and specify clearly the problem that you stated. For your study purposes, I included a classification of the problems and their difficulties:

- Warm-up: these are require that you understand some definition.
- *: they indicate some medium level of difficulty. You should spend longer in these problems.
- ** indicates that the problem is challenging, they will require more time than the other problems.

The complexity from ** and * is the level we would like you to master in this course. Questions highlighted with **OPTIONAL** are not necessary to get full credit for the homework, but I highly encourage you to solve them for your learning process.

In each of the solutions you **SHOULD JUSTIFY COMPLETELY YOUR ANSWER** with a proof or a counter-example. No credit is granted to solutions without justification.

1 Mandatory Problems

Definition 1.1. Let S be a set and $\star : S \times S \mapsto S$ a binary operation.

- $(S, \star)$ has an identity element if there is $e \in S$ such that for every $s \in S$:

$$s \star e = e \star s = s. \quad (1)$$

- Assume that e is an identity element. We say that it has inverses if for every $s \in S$ there is $w \in S$ such that:

$$s \star w = w \star s = e. \quad (2)$$

In this case we say that w is the inverse of s .

1. **Problem 1: (0.5 point)** Consider the set $A = \{3, 9, 15, 21\}$ together with the following operation:

$$x \star y := (xy \mod 24).$$

Determine whether the following properties hold in $(A, \star)$.

- a) $\star$ is associative,
 - b) $\star$ has an identity element. If so, does it have inverses? i.e. if e is the identity for every $x \in A$ is there a $y \in A$ such that $x \star y = y \star x = e$?
 - c) $\star$ is commutative.
2. **Problem 2* (6.5 points): (Isomorphisms)**
 - a) (2 points) Let $P_n = \{a_0 + a_1x + \dots + a_nx^n \mid a_i \in \mathbb{R}\}$ the set of polynomials of degree n with coefficients in the real numbers. Consider $(P_n, +)$ with the usual sum of polynomials. And consider $\mathbb{R}^{n+1}$ with the usual sum of vectors. Are $(P_n, +)$ and $(\mathbb{R}^{n+1}, +)$ isomorphic? Justify properly your answer.
 - b) (2.5 points) Consider the set $\text{GL}_2(\mathbb{R})$ of invertible 2×2 matrices over the real numbers.
 - i. Convince yourself that $\text{GL}_2(\mathbb{R})$ together with matrix multiplication has an identity and inverses.
 - ii. (0.5 point) Is the operation commutative? Provide a proof or a counterexample.
 - iii. (2 points) Consider the map $\det: \text{GL}_2(\mathbb{R}) \rightarrow \mathbb{R} \setminus \{0\}$. Is $\det$ an isomorphism?
 - c) (2 points) Let $f: \mathbb{Q} \mapsto \mathbb{Q}$ defined as $f(x) = 3x - 1$. Prove that it is a bijection. And define $\star$ an operation in $\mathbb{Q}$ such that f is an isomorphism mapping in:
 - $(\mathbb{Q}, \cdot)$ onto $(\mathbb{Q}, \star)$,
 - $(\mathbb{Q}, \star)$ onto $(\mathbb{Q}, \cdot)$.

In each case describe the identity element of $\star$ in $\mathbb{Q}$.

3. **Problem 3*: (3 points) (Structural properties)**

- a) (2 point) Show that the property of having an identity element and inverses is a structural property. This is:
If $(S, \star)$ has an identity element and every element has an inverse, then if $(S', \star')$ is isomorphic to $(S, \star)$ then $(S', \star')$ also has an identity element and inverses.

- b) (1 point) Are $(GL_n(\mathbb{R}), \cdot)$ and $(M_{n \times n}(\mathbb{R}), \cdot)$ isomorphic? Where $GL_n(\mathbb{R})$ denotes the $n \times n$ matrices with real entries whose determinant is non-zero, while $M_{n \times n}(\mathbb{R})$ are the $n \times n$ matrices of real entries.

2 Optional problems

I recommend the problems in the book:

1. Section (Isomorphism of binary structures) 12, 14 and 17.
2. Consider problem 2, are $(P_n, +)$ and $(\mathbb{R}^n, +)$ isomorphic? (This is harder: Hint use Zorn's Lemma and generating sets).